

Chat Insights

Session 3

Embeddings · vector DB vs traditional DB · Qdrant payloads · RAG Q&A

This document pulls together the sharpest and most broadly useful questions from the live chat during Session 3, along with the answers given by the mentors and the team. Personal logistics questions, roll call, reactions, and technical glitch chatter have been left out. This is a companion to the Session Notes, not a replacement, refer back to the Session Notes for the full explanation of any concept mentioned here in shorthand.

1. Fundamentals Clarifications: Embeddings and Tensors

Asked by	Question	Answer or takeaway
Kush Singh	What does contextual embedding mean in terms of a sentence or a document embedding, versus a word embedding?	Ashish (Team Codebasics): contextual means the embedding represents the meaning of the entire text in the context of the other words in that text, rather than representing each word independently. It captures the relationship and context.
Renganathan M	Does the embedding technique from the last session use the transformer approach? Doesn't running a transformer require massive infrastructure?	Sid (Team Codebasics): yes, it used the transformer approach. Resource needs scale with model size, all-MiniLM-L6-v2 is only about 22.7 million parameters and around 90 MB of memory. Large models have parameters in the trillions and need terabytes of compute.
Shubradip Ghosh	Changing the embedding model can make old and new vectors incompatible. How do we handle that?	Krishna (Team Codebasics): if you change the embedding model, you need to re-embed your documents using the new model. Old and new embeddings live in different vector spaces, so do not mix them in the same vector DB.
Abhishek Onkar	Does a vector database come with one fixed similarity type, or is it configurable per collection?	Syed Mizbahuddin (peer): the embedding model is trained on one strategy, but a database like Qdrant lets you choose a different distance strategy for each collection.

2. Vector Databases and Qdrant Q&A

Asked by	Question	Answer or takeaway
Bharath Kumar	Won't the vector database become huge if we store all columns along with the vector embeddings?	Krishna: we do care about other things like metadata, because it helps in filtering our data, so it earns its place in the payload.
Abhishek Onkar	If an RDBMS column can store a vector and support efficient similarity search, is a dedicated vector DB redundant?	Dhaval Patel (Team Codebasics): your assumption is right that vectors can live in an RDBMS, that's exactly why traditional databases like Postgres are adding vector support (pgVector). Vector DBs are not redundant, they still provide more efficient search and other benefits.
Bharath Kumar	Are vector databases a replacement for traditional databases?	Krishna: no. RDBMS remains the source of truth for structured data, the vector DB stores embeddings plus only the metadata needed for semantic retrieval.
Renganathan M	Where does the Qdrant database actually live when we run it in Colab, and where in production?	Krishna: in Colab, Qdrant runs in embedded (local) mode inside your Python process, the /tmp path is just where it stores collection files, and it is deleted when the runtime ends unless you download it.
Rajender S	In a production environment, where is Qdrant typically hosted?	Krishna: Qdrant Cloud (managed), or self hosted on Docker or Kubernetes on AWS, GCP, Azure, and similar.
Abhay Chauhan	How are fields inside a payload indexed for faster filtering?	Ashish: using metadata, similar to a database index.
Akhilesh	Can we have duplicate values in an ID?	Krishna: no, the ID has to be unique.

3. Cost, Sizing, and Production Q&A

Asked by	Question	Answer or takeaway
Pradeep Deshmukh	What does the formula "100000 * 384 * 4" for sizing a vector store actually mean?	Sid: number of records times dimension times precision bytes. By default embeddings are stored as FP32, which is 4 bytes per value.
Kamal	Where can we find the precision details for a given model?	Sid: FP16, FP8, and FP32 are the common floating point datatypes, by default it is FP32 unless the model or store says otherwise.
Pradeep Deshmukh	For this session's example (384 dim, 100K records), what kind of machine would that actually need?	Sid: with a 384 dimension embedding, a 4 GB RAM, 2 vCPU machine would be comfortable.

Key Insight (from Sandip)

score_threshold is really important to avoid hallucination. If you don't set one, the query will still return results under whatever limit you asked for, even if none of them are actually relevant.

4. RAG Q&A

Asked by	Question	Answer or takeaway
Murthy Rajender	Each chunk is split based on a fixed length, won't that lose context between chunks?	Krishna: we usually use chunk overlap, where consecutive chunks share some text so important information isn't cut off between them.
Himesh Sarkari	What if the answer to a question actually spans more than one chunk?	Krishna: we send the top several chunks (for example the top 5) to the LLM together, rather than only the single best match.
renjith r	How will the LLM have full context if we are only giving it the query results?	Krishna: the query results already contain all the relevant context needed, that's exactly the point of RAG, sending only what's relevant instead of everything.
Abhisekh	Could you briefly explain for idx, (chunk, embedding) in enumerate(zip(chunks, embeddings))?	Krishna: zip(chunks, embeddings) pairs the two lists item by item, for example (A, E1) and (B, E2). enumerate(...) then adds a running index, giving (0, (A, E1)), (1, (B, E2)), and so on.
Ankit Garg	How is an existing embedding updated if the source data (a PDF or log file) changes?	Krishna: a webhook or similar signal tells the system the document changed, which triggers a re-ingestion pipeline. Since documents have their own IDs in the payload, you delete the old chunks for that document ID and re-chunk and re-embed the new version.

Key Insight (from Ashish, Team Codebasics)

The core of retrieval in one line: query becomes embedding vectors, each document chunk becomes embedding vectors, then you compare vectors using cosine similarity (or another method), rank the results, and use the top ones to generate the answer.

5. Standout Participant Insights

Key Insight (from akhil)

Static embeddings give you a dictionary of meanings. Contextual embeddings give you a meaning that adapts to the sentence the word is actually sitting in, which is why contextual embeddings, and the transformer architectures behind them, became the backbone of modern NLP and LLMs.

Key Insight (from Gopinath)

On why the hands on notebook produced 51 chunks of 384 dimensions: 51 is the number of chunks the document was split into, and 384 is the embedding dimension of the model, so you end up with 51 separate vectors, each 384 numbers long, one per chunk.

Key Insight (from Ashu Swami)

RAG is an Iron Man suit, and the LLM is Tony Stark. The base model already has real intelligence and language ability on its own, RAG just hands it extra power and up to date information it wouldn't otherwise have.

Key Insight (from bharath vamsi)

Answering a peer who asked whether they could skip ahead to the Transformers lecture without the earlier deep learning material: you can understand transformers a lot better if you go a step back and start from sequence models first, rather than jumping straight in.

6. Curriculum and Pacing

Asked by	Question	Answer or takeaway
Ashu Swami	For the recorded bootcamp material, how should learners pace themselves alongside the live sessions?	Krishna: it's okay to watch the recorded material in parallel with the live sessions, you don't need to know everything at once, and you can catch up on recordings across the coming week.
Sanjeev Rohila	Could all the session notebooks be kept in one centralized repo instead of shared one at a time?	Ashish: files are deliberately not shared all at once, so concepts are unravelled one topic at a time instead of overwhelming learners. All notebooks for a session are shared afterward, in Discord, in a single combined file.

7. Vector Database Tools the Cohort Mentioned

Tool	Context from chat
Qdrant	The vector DB used for this session's hands on exercises
Chroma / ChromaDB	Mentioned by several learners as an alternative; the mentor later noted in session that Chroma is mainly used for academic and course purposes
Pinecone	Mentioned as a popular managed vector DB option
Milvus	Mentioned as an open source vector DB alternative
pgvector (Postgres)	Raised in the context of adding vector search directly inside a relational database
Weaviate	Mentioned as another vector DB option by the cohort
Azure AI Search	Raised as Azure's own vector search offering for teams already on that cloud
Cosmos DB (vCore)	Mentioned by a learner as their own choice for storing embeddings